THE VACCINE LIST

Vaccines approved for Emergency Use Authorisation (EUA) by India

CORBEVAX
- Biological E Limited
- Protein sub-unit
- 1 country

COVOVAX
- Serum Institute of India
- Protein sub-unit
- 3 countries

ZYCOV D
- Zydus Cadila
- DNA
- 3 countries

MRNA1273
- Moderna
- RNA
- 85 countries

SPUTNIK V
- Gamaleya
- Non-replicating viral vector
- 74 countries

AD.26.COV2.S
- Janssen (Johnson & Johnson)
- Non-replicating viral vector
- 100 countries

AZD122
- Oxford Astra Zeneca
- Non-replicating viral vector
- 134 countries

COVISHIELD
- Serum Institute of India
- Non-replicating viral vector
- 47 countries

COVAXIN
- Bharat Biotech
- Inactivated
- 13 countries

- Manufacturer
- Type
- No. of countries where it is approved

Only Covishield and Covaxin are being used for the booster doses

Graphic by TANMOY CHAKRABORTY

virus. Vaccination is not intended to load up the body with antibodies. It is intended to sensitise the immune system; once sensitised, it will fight back any time it is asked to do so.

VACCINE PROSPECTS

Vaccine-makers maintain that inoculation is the only immunity and are, therefore, working on newer iterations for emerging situations depending on how the virus mutates. They rule out developing a single vaccine to treat all variants. Bharat Biotech, awaiting the outcome of studies, feels that a three-dose vaccine will be effective against Omicron. "The depth of science is bettering innovation in India," says CMD Krishna Ella, who also plans to register a patent. "We are close to unveiling a nasal vaccine that will provide immunity and also help break transmission." He believes that apart from Covid-19 turning endemic, variants will continue to exist and be treated like influenza, with one or the other variant playing up in one or the other region every year.

Last month, India granted EUA to two new Covid vaccines—Corbevax and Covovax. Corbevax was developed by the Hyderabad-based biopharmaceutical firm Biological E, along with Baylor College of Medicine and Dynavax Technologies in the US. It uses a recombinant protein-based technology, one that has been around since the 1980s and is used in the Hepatitis B vaccine. Containing a protein of the coronavirus, it arms the body to recognise the virus and launch an immune response against it. It is unlike Covaxin, which contains the whole virus but in an inactivated form. And since this vaccine contains only a part of the virus, an adjuvant is added to make the immune response stronger.

Covovax, also a recombinant-based vaccine, developed by the US-based Novavax, will be produced by the Pune-based Serum Institute of India. To develop it, researchers isolated the part of the genetic sequence of the virus that produces the spike protein and introduced it in baculoviruses or pathogens that attack insects, instead of using yeast as Corbevax does. Like all vaccines, Covovax tricks the immune system into believing it is under attack from the real virus.

Besides the vaccines, Molnupiravir—an anti-viral that its makers say can help with moderate Covid—has also been approved and 13 companies are set to manufacture the drug. But the Indian Council of Medical Research says the drug is unsafe. To achieve complete vaccination by mid-2022, production will also be ramped up by getting manufacturers to use available unutilised production spaces like the Integrated Vaccine Complex at Chengalpattu in Tamil Nadu.

Meanwhile, US Army researchers are developing a new "super vaccine" that can tackle the virus in all its mutations, as well as SARS-CoV-1, which was first detected in 2003. Known as the Spike Ferritin Nanoparticle (SpFN) vaccine, its creators say the shot generates strong binding antibody responses against the receptor-binding domain and spike proteins of all variants of concern. The US Army's strategy is to develop a 'pan-coronavirus' vaccine technology that could potentially offer safe, effective and durable protection against multiple coronavirus strains and species.

Equally critical is developing a robust public health surveillance system. Epidemiologists say every state must have genomic surveillance in place, with a comprehensive sampling strategy to track any change. This will require regular testing a fixed number of samples from each part of every state. Some experts warn that new Covid-19 variants will keep appearing until the entire world has been vaccinated against the virus. They argue that sharing vaccines is not just an act of altruism but also of pragmatism. World Health Organization data reveals that the global Covid-19 toll in 2021, as of December 29, was 5,411,759, which exceeded the combined toll from HIV/ AIDS, tuberculosis and malaria in 2020. Vaccination is the only way forward. ■